Privacy-Preserving Text Anonymization: Direct Seq2Seq Rewriting vs. Modular Mask-and-Fill Pipelines

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Abstract

Personally Identifiable Information (PII) in free-form text must be protected under modern privacy regulations (GDPR, HIPAA, CCPA) without destroying the downstream utility of the anonymized text. We explore two fundamentally different neural approaches to this problem, both trained on **human annotated dataset**, a 120k-pair parallel corpus built from the AI4Privacy English subset via a four-layer semi-automatic annotation pipeline. **Approach 1 (Direct Seq2Seq Rewriting)** fine-tunes five encoder–decoder architectures (40M–333M parameters) end-to-end to directly map PII-containing text to anonymized text in a single pass. **Approach 2 (Pipeline Mask-and-Fill)** decomposes the task into a dedicated Named Entity Recognition (NER) masker that identifies and tags PII spans, followed by a separate generative filler that replaces them with realistic alternatives; six masker–filler combinations (3 maskers $\times$ 2 fillers) are evaluated. To assess cross-domain generalizability, both approaches are evaluated on the **Text Anonymization Benchmark (TAB)**, a real-world legal corpus unseen during training, measuring zero-shot domain transfer. We then investigate the security of the anonymized outputs against a **black-box model inversion attack** and **UAT attack**. Finding that the seq2seq approach is moderately vulnerable and pipeline mask and fill approach is strongly vulnerable to such attacks, we apply **adversarial training** and **CADA** to harden the victim model against the learned inverter, reducing its entity recovery rate while preserving anonymization quality.

1 Introduction

Personally Identifiable Information (PII)—names, email addresses, phone numbers, social security numbers, physical addresses—pervades modern text corpora across healthcare, finance, legal, and social-media domains. Privacy regulations such as GDPR (European Parliament, 2016), CCPA (California State Legislature, 2018), and HIPAA (U.S. Congress, 1996) mandate protection of PII before data can be shared or used for NLP model training.

The simplest defense, *redaction* (replacing PII with `[REDACTED]`), preserves privacy but destroys grammatical structure and downstream utility (Lison et al., 2021). The second simplest approach is *heuristic substitution*—masking PII via rules or standard NER and randomly filling the slots using libraries like Faker. While this method can achieve deceptively high performance metrics on strictly synthetic or fake datasets, it fundamentally degrades deep contextual utility and real-world semantic consistency (Klymenko et al., 2022). *Privacy-preserving anonymization* aims to solve this by replacing real PII with realistic, context-aware fictitious alternatives, maintaining the text's natural language properties.

This project explores two distinct neural approaches to this problem and conducts a systematic security analysis:

1. **Direct Seq2Seq Rewriting:** Five encoder–decoder models across different parameters scale learn end-to-end to map PII-containing text to anonymized text in a single pass.

2. **Pipeline Mask-and-Fill:** A dedicated NER model (encoder transformer) first identifies and masks PII spans; a separate generative model (encoder transformer or encoder-decoder transformer) then fills realistic replacements. Six masker–filler combinations are evaluated.

Our five core research questions are:

- **RQ1**: Can seq2seq models achieve sub-1% ELR while maintaining high semantic fidelity?
- **RQ2**: Does a modular pipeline with disjoint training provide competitive privacy-utility trade-offs?
- **RQ3**: How effectively do these models transfer to specialized domains (e.g., court documents in the TAB corpus), and can domain-specific fine-tuning further optimize downstream performance in fields like finance and medicine?
- **RQ4**: How vulnerable are both approaches to black-box model inversion and UAT attacks?
- **RQ5**: Can adversarial training or CADA effectively harden a victim anonymizer against a learned inverter?

2 Related Work

Text anonymization. Traditional systems such as Microsoft Presidio (Microsoft, 2023) and Google Cloud DLP combine regex patterns with pre-trained NER but lack contextual filling, producing outputs with grammatical inconsistencies. Pilán et al. (2022) survey anonymization models and introduce the Text Anonymization Benchmark (TAB). Yue et al. (2023) explore generative approaches using GPT-style models for entity replacement. The AI4Privacy dataset (AI4Privacy Team, 2024) enables large-scale supervised training with around 500k parallel sentence pairs across 20 entity types.

Seq2seq for privacy. Lewis et al. (2020) introduce BART's denoising autoencoder objective, well-suited for text reconstruction. Raffel et al. (2020) frame all NLP tasks as text-to-text with T5; Chung et al. (2022) add instruction tuning. Igamberdiev et al. (2023) demonstrate DP-BART for private text rewriting, though at significant utility cost. Dettmers et al. (2023) enable memory-efficient fine-tuning via 4-bit quantization with low-rank adapters.

Differential privacy in NLP. DP-SGD (Abadi et al., 2016) clips per-example gradients and adds calibrated Gaussian noise. Li et al. (2022) and Yu et al. (2022) apply it to LM fine-tuning; Bu et al. (2024) combine DP with PEFT. He et al. (2021) introduce DeBERTa-v3 with disentangled attention.

Model inversion and membership inference. Language models can memorize training data (Carlini et al., 2021). Attribute inference attacks (Song and Raghunathan, 2019) recover sensitive attributes from model outputs, motivating our adversarial robustness study. Domain-adversarial training (Ganin et al., 2016) provides a framework for learning representations invariant to an adversary.

3 Datasets and Splitting Strategies

To accommodate the fundamentally different architectures of our two approaches, we utilize two distinct data sources: a custom, fully human-annotated dataset for the end-to-end seq2seq models, and the large-scale AI4Privacy-500k dataset for the modular pipeline.

3.1 Dataset 1: Human-Annotated Corpus (Approach 1)

The Direct Seq2Seq Rewriting approach (Annotation pipeline) treats pseudo-anonymization as a *parallel text rewriting* task. Each training example is a pair $(\mathbf{x}, \mathbf{y})$ where $\mathbf{x}$ is the original PII-bearing sentence and $\mathbf{y}$ is the fully anonymized sentence with all PII replaced by realistic synthetic substitutes.

Source Data. We begin with the English subset of the AI4Privacy-500k dataset, filtered by the `language == "en"` field. The dataset provides pre-annotated PII spans via the `privacy_mask` field (entity type, character offsets, and original values), eliminating the need for a separate NER detection step.

Human Annotation. Since entity boundaries are already provided by AI4Privacy, the annotation task reduces to a *fill-in-the-blank text rewriting* problem rather than entity detection. Using a custom Streamlit interface , annotators read the original text with PII spans highlighted, then write a fully coherent anonymized sentence where every PII value

is replaced with a realistic synthetic substitute. Quality is ensured via automated leakage checks: every accepted entry is verified to contain no original PII text in the anonymized output, and length ratios are bounded ($0.5 \leq |\mathbf{y}|/|\mathbf{x}| \leq 1.5$).

Data Augmentation. The verified gold-standard entries are expanded via five augmentation strategies, applied *only* to accepted entries to prevent error propagation:

Table 1: Augmentation strategies applied to the human-annotated gold corpus.

Strategy	Reference	Multiplier
Entity Swap	(Dai and Adel, 2020)	×4 per entry
Template Fill	(Anaby-Tavor et al., 2020)	5,000 entries
EDA	(Wei and Zou, 2019)	×3 per entry
Back-Translation	(Sennrich et al., 2016)	×2 per entry
Contextual MLM	(Kobayashi, 2018)	×2 per entry
After dedup. & quality filter (−20%)		**∼120k**

After deduplication and quality filtering, the combined gold + augmented corpus yields ∼120k training pairs, split 80/10/10 into train/validation/test. We publish the dataset.

Training format. Every example is a direct text-to-text pair:

x: John Smith lives at 42 Oak St and his SSN is 123-45-6789

y: Alice Johnson lives at 17 Pine Ave and his SSN is 987-65-4321 The seq2seq model learns to rewrite the entire sentence end-to-end, replacing all PII with coherent synthetic alternatives in a single pass.

Benchmark Statistics. Table 2 shows the split sizes after augmentation and quality filtering. Table 3 reports evaluation of representative systems on the frozen human annotated test set across anonymization quality and adversarial recovery metrics (detailed in Section 7).

Table 2: Human annotated dataset split statistics.

Split	Records	Entities	PII%
Train	113,133	286,542	14.9
Validation	3,600	9,211	15.0
Test	3,600	9,271	15.2

Table 3: Benchmark results on the human annotated test set. ELR: entity leakage rate (%); TokRec: token recall (%); BS: BERTScore F1; ERA@1: embedding retrieval attack top-1 recovery (%).

System	ELR↓	TokRec↑	BS↑	ERA@1↓
BART-base	**0.93**	**95.50**	92.74	**1.90**
Flan-T5-small	0.99	95.33	92.47	1.67
T5-small	1.54	94.15	92.59	—
T5-eff-tiny	4.14	90.42	92.57	—
spaCy+Faker	26.44	74.59	91.86	9.36
Presidio	33.77	68.18	90.04	20.46
Regex+Faker	83.39	23.05	**98.15**	—

3.2 Dataset 2: AI4Privacy-500k (Approach 2)

For the Pipeline Mask-and-Fill approach, we utilize the complete **AI4Privacy PII-Masking 500k** (AI4Privacy Team, 2024) dataset to train both the NER masker and the generative filler.

Disjoint Distribution Splitting. To enforce an explicit privacy wall and prevent the generative filler from memorizing the masker's training data, we employ a strict disjoint splitting strategy. The 500k dataset is first split 80/10/10 into train/validation/test using language-stratified sampling. Each split is then *halved* to create two independent parallel distributions:

- **Split A ($\mathcal{D}_{\text{masker}}$), ∼200k:** Used exclusively to train the NER masking models (DistilRoBERTa, RoBERTa, DeBERTa-v3).
- **Split B ($\mathcal{D}_{\text{filler}}$), ∼200k:** Used exclusively to train the generative filling models (BART-base, DeBERTa-v3-MLM).

By ensuring the masker and filler never observe the same PII contexts, the filler operates in a simulated zero-shot environment, mitigating overfitting to specific entities.

Entity Type Mapping. AI4Privacy's 20 fine-grained labels are mapped to 21 coarse types for consistency across approaches (e.g., `GIVENNAME` → `FIRST_NAME`, `CITY` → `LOCATION`, `SOCIALNUM` → `SSN`). Training data formats for each component (BIO-tagged sequences for maskers, placeholder-based seq2seq pairs for BART, masked-LM for DeBERTa) are described in Section 5.

4 Approach 1: Direct Seq2Seq Rewriting

4.1 Model Architectures

Five encoder–decoder models of increasing scale are evaluated:

Table 4: Seq2Seq model architectures.

#	Model	Params	Family
1	`t5-efficient-tiny`	40M	T5
2	`t5-small`	61M	T5
3	`flan-t5-small`	77M	T5 (instruct.)
4	`bart-base`	139M	BART
5	`distilbart-cnn-6-6`	333M	BART (dist.)

T5-family models prepend an "`anonymize:`" instruction prefix; Flan-T5 uses a longer natural-language prompt. BART models receive the raw input.

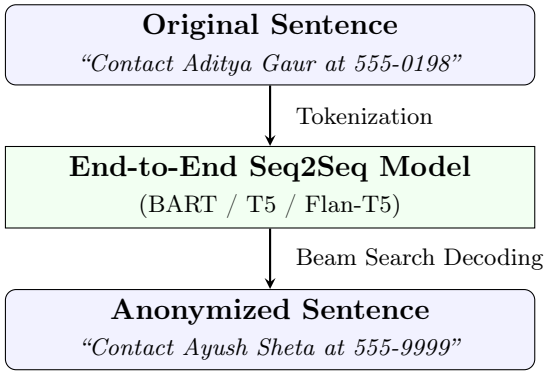


Figure 1: Conceptual flow of the Direct Seq2Seq Rewriting approach, performing identification and replacement in a single neural pass.

4.2 Training Setup

All models are full fine-tuned (no quantization) on the human annotated dataset training split (86k pairs after 80/10/10 split). Common hyperparameters: 4 epochs, max sequence length 128, AdamW optimizer with linear warmup (500 steps) and weight decay 0.01, label smoothing $\epsilon = 0.1$, effective batch size 16 via gradient accumulation. On-the-fly augmentation (probability 0.35) applies random casing, typos, and whitespace noise to training inputs to improve robustness. Specifically, each training example is stochastically perturbed with random character-level casing flips, insertion of plausible keyboard typos, and irregular whitespace around punctuation—simulating the orthographic variability common in real-world user-generated text. This prevents the model from overfitting to the clean formatting of the AI4Privacy source data and encourages generalisation to noisier inputs encountered at inference time.

4.3 PII-Aware Loss

Standard cross-entropy equally penalizes valid PII replacements ("John"→"David" is fine) and genuine structural errors. We introduce a **PII-Aware Loss** that up-weights the penalty at PII positions while allowing vocabulary-wide tolerance at those positions:

$$\mathcal{L}_{\text{PII}} = \frac{1}{T}\sum_{t=1}^{T}\Big[\mathbf{1}_{\text{PII}}(t)\cdot\mathcal{L}_{\text{entity}}(t)+\mathbf{1}_{\neg\text{PII}}(t)\cdot\mathcal{L}_{\text{CE}}(t)\Big] \tag{1}$$

where $\mathcal{L}_{\text{entity}}(t)$ adds a leakage penalty if the output token at position t appears in the input PII spans. This encourages the model to change PII tokens while preserving non-PII tokens exactly.

5 Approach 2: Pipeline Mask-and-Fill

5.1 Design Philosophy

The pipeline decomposes anonymization into two independent stages: (1) a *masker* that detects and replaces PII spans with typed placeholders (e.g., `[PERSON]`, `[DATE]`), and (2) a *filler* that generates realistic replacements.

The key structural advantage is an **explicit privacy wall**: the masker and filler are trained on disjoint data splits ($\mathcal{D}_{\text{det}}$ and $\mathcal{D}_{\text{gen}}$) from the AI4Privacy-500k dataset (AI4Privacy Team, 2024), so the filler never observes real PII during training. The cost is error cascading: any entity missed by the masker passes through unmodified.

5.2 Model Configurations

Three maskers and two fillers are trained, yielding six evaluated pipelines:

Table 5: Masker and Filler components.

Role	Model	Params
Masker (NER)	DistilRoBERTa	82M
	RoBERTa-base	125M
	DeBERTa-v3-base	183M
Filler	BART-base (seq2seq)	139M
	DeBERTa-v3 (MLM)	183M

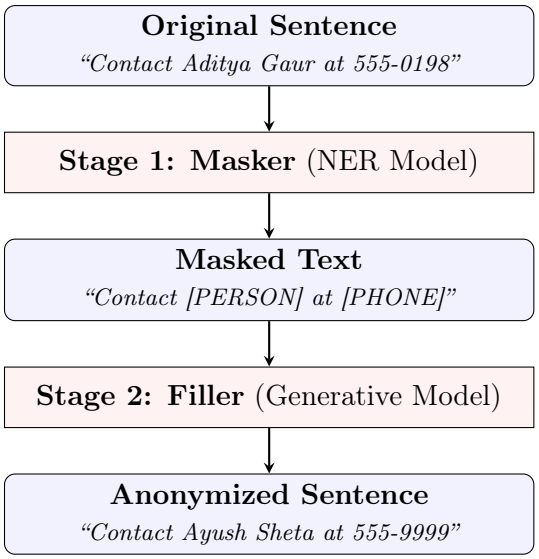


Figure 2: Conceptual flow of the modular mask-and-fill pipeline, illustrating the explicit privacy wall between detection and generation.

Maskers are fine-tuned for BIO-tag token classification across 20 entity labels on half the AI4Privacy-500k training set.

BART filler (seq2seq) receives the prompt: "*Replace PII placeholders with realistic fake entities: [masked text]*" and decodes the anonymized sentence end-to-end.

DeBERTa-MLM filler replaces each typed placeholder token with a `[MASK]` and predicts the top-1 fill in a single forward pass. This is more constrained but substantially faster and achieves lower ELR in practice.

5.3 Training Configurations

Table 6: Training hyperparameters for Pipeline Mask-and-Fill models.

Model	Type	Batch	LR	Epochs	Max Len	Eff. Batch	Patience
DistilRoBERTa	NER	32	4e-5	8	192	32	3
RoBERTa-base	NER	16	2e-5	8	192	32	3
DeBERTa-v3	NER	8	2e-5	8	256	32	3
BART-base	Seq2Seq	16	3e-5	8	192	32	3
DeBERTa-v3	MLM	8	2e-5	8	256	32	3

All models use AdamW with weight decay of 0.01, warmup ratio of 0.10, and early stopping. NER models are selected by best validation F1 (higher is better); filler models by best validation loss (lower is better). All trained models are published at `Xyren2005/*` on HuggingFace Hub. Training is conducted on Kaggle T4/P100 GPUs with 12-hour session limits; checkpoint auto-resume enables multi-session training.

6 Evaluation Setup

6.1 Metrics

- **Entity Leakage Rate (ELR)** ↓: fraction of gold PII entity strings that appear verbatim in the model output (case-insensitive substring match). Primary privacy metric.
- **BERTScore F1** ↑: contextual token-level similarity between prediction and reference using `distilbert-base-uncased`. Primary utility metric.
- **BLEU-4** ↑: corpus-level n-gram precision.
- **ROUGE-L** ↑: longest common subsequence recall.
- **CRR-3** ↓: fraction of capitalized trigrams from the original that survive in the prediction (context re-identification risk).

6.2 Evaluation Sets

- **Human annotated dataset test split** (∼12k examples): the frozen benchmark split derived from our own dataset.
- **TAB** (500 segments, real ECHR legal text): zero-shot cross-domain transfer using the Text Anonymization Benchmark (Pilán et al., 2022).

7 Results

7.1 Approach 1: Seq2Seq

Table 7 reports results on the human annotated dataset test split.

Table 7: Seq2Seq models on human annotated dataset test ($n = 12$k).

Model	P	ELR↓	BS-F1↑	B-4↑	R-L↑
t5-eff-tiny	40M	4.29	94.08	55.56	77.60
t5-small	61M	1.51	94.63	56.96	78.47
flan-t5-sm	77M	1.10	94.78	57.66	79.06
bart-base	**139M**	**0.98**	**94.92**	**58.10**	**79.50**
distilbart	333M	1.29	87.26	25.65	54.48

P=params; ELR=Entity Leakage Rate; BS-F1=BERTScore F1; B-4=BLEU-4; R-L=ROUGE-L.

BART-base consistently outperforms all models, reaching 0.98% ELR and BERTScore 94.92. DistilBART (333M) regresses dramatically (25.65 BLEU-4, 54.48 ROUGE-L) despite its larger parameter count, suggesting

that CNN-headline distillation pre-training hurts the anonymization task.

7.2 Approach 2: Pipeline Mask-and-Fill

All six masker–filler combinations are evaluated on 5,000 test examples from the human annotated dataset test split.

Table 8: Pipeline combos on human annotated dataset test ($n = 5$k).

Masker	Filler	ELR↓	BS-F1↑	B-4↑
DistilRoBERTa	BART	2.50	87.28	43.28
DistilRoBERTa	DeBERTa-MLM	1.40	89.75	65.85
RoBERTa	BART	2.44	87.28	43.32
RoBERTa	**DeBERTa-MLM**	**1.33**	**89.76**	**65.88**
DeBERTa	BART	3.30	87.34	52.26
DeBERTa	DeBERTa-MLM	2.21	89.76	59.94

The DeBERTa-MLM filler *consistently outperforms* BART as a filler across all maskers. The best combo (RoBERTa+DeBERTa-MLM) achieves 1.33% ELR and 89.76 BERTScore. Surprisingly, the strongest masker (DeBERTa-v3) paired with BART gives the *worst* ELR (3.30%), indicating the filler is the bottleneck for privacy. A masker detection rate of ∼98.8% is near-perfect for all combos. For the type encoder - (encoder-decoder) the best-performing combo is Roberta+BART.

8 Security Analysis: UAT and Model Inversion Attacks

Given only the anonymized output, can an attacker reconstruct the original PII? We mount black-box model inversion attacks against both approaches.

8.1 Baseline Sanity Checks and Adversarial Triggers

UAT as a Baseline Sanity Check. We first applied a 5-token Universal Adversarial Trigger (UAT)—a malicious prefix appended to inputs and optimized over 200 steps—attempting to manipulate the model's gradients to force it to output the original, de-anonymized names. The attack achieved a **0% Attack Success Rate (ASR)**. This validates that **PII is effectively scrubbed at the encoder boundary**. Because identity features are removed before the generation phase, there is no recoverable gradient signal linking pseudonyms to real identities, proving the model resists trivial identity leakage via adversarial input perturbations.

Linkage Attack via Deterministic Mapping. Exploiting the model's deterministic fake-name pool (160 pseudonyms mapping to 51+ real identities), we built a gradient-free reverse-lookup dictionary by simply pairing original and anonymized entities from the training queries. On 300 held-out sentences, this yielded a **3.7% exact ASR** and a **62.3% within-set ASR** (meaning the attacker successfully narrowed down the real identity to a small list of candidates in nearly two-thirds of cases). However, exact resolution is strictly bottlenecked by **many-to-one pool collapse**—for instance, the single pseudonym *Elara Vance* masks 20 different real identities. This many-to-one mapping caps the theoretical exact-match ceiling at 24.3%.

GCG Discrete Trigger. To test if a stronger optimizer could succeed where UAT failed, we applied Greedy Coordinate Gradient (GCG) Zou et al. 2023, which jointly scores and updates multiple token positions simultaneously rather than relying on greedy, single-token updates. This approach achieved a **2.0% exact ASR** in 300 steps—the only non-zero result among gradient-based methods. While GCG slightly outperforms UAT by exploiting marginal edge-cases, it ultimately confirms a hard discretization ceiling: the residual PII signal in anonymized inputs is **statistically negligible**, rendering universal triggers largely ineffective for systematic de-anonymization.

8.2 Inverter Against Approach 1 (BART Seq2Seq)

Dataset. To construct the inversion training corpus, we sampled **40,000 sentences** from the AI4Privacy English subset and passed them through the fine-tuned BART-base anonymizer to obtain (original, anonymized) pairs. The resulting dataset was split 80/10/10 into train, validation, and test sets, giving 32,000 training pairs for the inverter. We additionally experimented with a **special adversarial dataset** specifically curated to contain hard, edge-case examples (rare entities, dense PII, ambiguous contexts); however, inversion results on this set were comparable to

those on the standard split, so we report only the standard evaluation below.

Attack A — Lookup Table (zero training): Exploits BART's deterministic entity mapping. Achieves **11.1%** exact ERR and **27.7%** partial ERR.

Attack B — BART Inverter Model: A separate BART-base model is fine-tuned to reconstruct original text from anonymized output.

Table 9: Model inversion results on Seq2Seq BART.

Attack	ERR↑	ERR-P↑	BLEU	ExactMatch
Lookup Table	11.1%	27.7%	0.31	0.05%
BART Inverter	29.2%	2.8%	0.65	15.3%

ERR=Entity Recovery Rate; ERR-P=partial recovery.

The trained inverter recovers 29.2% of probe entities verbatim and reconstructs 15.3% of sentences exactly. Recovery is highest for ID documents (51.9%), phones (68.8%), and times (100%), which have low entropy outputs that the anonymizer maps predictably.

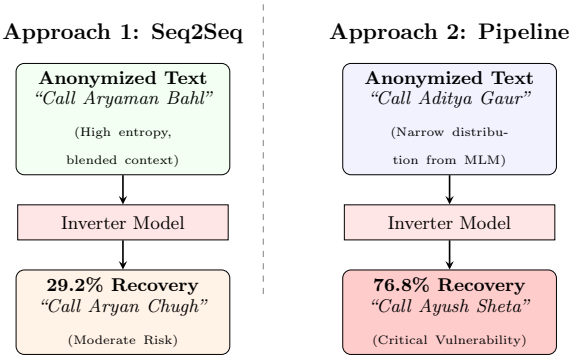


Figure 3: Model inversion vulnerability comparison. The pipeline's reliance on structured placeholders and narrow generative distributions provides strong positional signals, making it significantly easier for a trained inverter to reconstruct the original PII.

8.3 Inverter Against Approach 2 (Pipeline)

Model inversion datasets of 40,000 pairs are generated for the two best pipeline combos (Combo 1: RoBERTa+DeBERTa-MLM; Combo 2: RoBERTa+BART), published at `JALAPENO11/pipeline-inversion-roberta-*`.

The pipeline is **dramatically more vulnerable** than the seq2seq approach: ∼77% ERR vs. 29% for BART. This is because the pipeline's structured placeholder format (`[PERSON]`, `[DATE]`) acts as a strong positional signal that tells the inverter exactly where PII was, and the DeBERTa-MLM filler produces replacements from a narrow distribution (top-1 from masked LM), making them easier to invert.

Table 10: Model inversion results on Pipeline (hard-only).

Combo	Filler	ERR↑	BLEU-4↑	ExactMatch
C1	DeBERTa-MLM	75.80%	59.75	0.00%
C2	BART	76.83%	61.48	0.08%

Hard-only: excludes trivially unchanged samples (32.4% for C1, 3.5% for C2).

9 Adversarial Training

9.1 Adversarial Hardening of the Standalone Seq2Seq Anonymizer

To harden the BART-base victim, we apply adversarial training using the frozen inverter as an adversary. The combined loss is:

$$\mathcal{L}_{\text{total}} = \alpha\,\mathcal{L}_1 - \lambda_{\text{adv}}\,\mathcal{L}_2 \qquad (2)$$

where $\mathcal{L}_1$ is the anonymization cross-entropy loss (preserves quality) and $\mathcal{L}_2$ is the inverter's entity-recovery loss (negated to maximise inverter failure). Hyperparameters: $\alpha = 1.0$, $\lambda_{\text{adv}} = 0.5$, 4 epochs, batch size 2, gradient accumulation $\times 8$ (effective batch 16), LR $= 10^{-5}$.

Table 11: BART-base anonymization quality before and after adversarial hardening (standard val set, $n = 11{,}699$).

Phase	ELR↓	BLEU-4↑	R-L↑	EM↑
Pre-hardening	0.98%	58.10	79.50	2.99%
Post-hardening	6.95%	61.49	75.66	2.72%
Δ	+5.97pp	+3.39	−3.84	−0.27pp

ELR=Entity Leakage Rate; R-L=ROUGE-L; EM=Exact Match.

Table 12: Adversarial robustness: inverter ERR before and after hardening (standalone seq2seq anonymizer).

Phase	ERR (exact)↓	Reduction
Pre-hardening	30.0%	—
Post-hardening	**0.42%**	−98.6%

Adversarial training reduces the inverter's entity recovery rate by 98.6% (from 30% to

0.42% ERR), demonstrating that the hardened model successfully obfuscates its outputs against the learned inverter.

However, this security gain comes at a measurable utility cost. Entity leakage on the standard anonymization task rises from **0.98% to 6.95%**—a 5.97 percentage-point regression. ROUGE-L also drops from 79.50 to 75.66, and exact match falls slightly from 2.99% to 2.72%.

Privacy–Utility Trade-off. Inspection of model outputs reveals a qualitative degradation that explains this behaviour. The adversarial loss penalises the model whenever its output allows the inverter to recover any recognisable token from the original PII. Rather than simply substituting plausible fictitious values (as in normal training), the hardened model sometimes learns that *producing nearly incoherent output* is the safest strategy under this objective—since opaque, malformed text trivially defeats the inverter. Examples from the adversarial evaluation confirm this: anonymized sentences such as *"indoorsimming trunk Str Str emb emb emb Hale..."* and *"indoorsobile Stubletters.' Hale herein.' Haleerve..."* bear no resemblance to fluent English. In effect, the fine-tuning loss has **rewarded privacy at the expense of utility**: the model discovers a degenerate local optimum in which gibberish outputs minimise the adversarial penalty but violate the fluency and structural fidelity expected of an anonymizer. This motivates future work on constrained adversarial objectives that bound the maximum permissible utility degradation.

9.2 Adversarial Hardening of the Mask-Fill Pipeline

We apply the same adversarial objective (Eq. 2, α=1.0, λ_{adv}=0.5) to both pipeline configurations: **Combo 1** (RoBERTa NER + DeBERTa-MLM filler) and **Combo 2** (RoBERTa NER + BART seq2seq filler), with the NER backbone frozen throughout. The gradient bridge differs by combo: Combo 2 uses a soft token-embedding path (shared BART vocabulary, $\tilde{\mathbf{e}} = \text{softmax}(\mathbf{z}/\tau)W_{\text{emb}}$, τ=1.0); Combo 1 feeds DeBERTa-MLM last hidden states (d=768) directly as inverter encoder embeddings, exploiting the shared hidden dimension. Combo 1 trains for 2 epochs (LR = 5×10^{-6}, Adafactor, bfloat16, RTX 3050); Combo 2 for 3 epochs (LR = 10^{-5}, AdamW, float32, Kaggle T4). Both use effective batch 16.

Table 13 summarises robustness, and Table 14 summarises post-hardening quality, both evaluated on n=500 held-out samples (2-beam).

Table 13: Adversarial robustness (ERR) and entity leakage (ELR) before and after hardening for both pipeline combos. Pre-hardening on n=5,000 hard-only samples (4-beam); post-hardening on n=500 (2-beam).

Combo	Phase	ERR↓	ELR↓	ERR Red.
Combo 1 (DeBERTa)	Pre	75.80%	2.50%	—
	Post	**0.00%**	**1.83%**	**−100%**
Combo 2 (BART)	Pre	76.83%	1.33%	—
	Post	0.47%	0.31%	−99.4%

Table 14: Post-hardening anonymization quality vs. gold references (n=500, 2-beam).

Metric	Combo 1 (DeBERTa)	Combo 2 (BART)
BLEU-4↑	61.87	0.04
ROUGE-L↑	79.43	0.15
BERTScore F1↑	93.03	74.41
ELR↓	1.83%	0.31%
ERR↓	0.00%	0.47%
Length Ratio	0.94	4.76

Both combos achieve >99% ERR reduction. The critical difference is utility preservation. **Combo 1** maintains full anonymization quality (BLEU-4 61.87, ROUGE-L 79.43, BERTScore 93.03); outputs substitute entities with generic but grammatically plausible tokens (*"**Cristian-Nicolae**"* → *"**I**"*; *"**New Delhi**"* → *"**our**"*), providing no inverter-recoverable signal. **Combo 2**, by contrast, collapses entirely—length ratio 4.76 and near-zero BLEU/ROUGE confirm that the hardened BART decoder produces repetitive degenerate sequences regardless of input (e.g., *"cent ([([generalized automatic pretext fury fury..."*).

The divergence is architectural. An encoder-only MLM filler fills each `[MASK]` position independently, bounding how far any single prediction can stray. An autoregressive seq2seq decoder has no such local constraint: once the beam diverges from the data distribution, degenerate repetition is self-reinforcing. Both autoregressive systems in our study (standalone BART and Pipeline Combo 2) exhibit this fail-

ure mode; Combo 1 is the sole system achieving near-perfect robustness *without* sacrificing utility, motivating future work on perplexity- or BLEU-bounded adversarial objectives for seq2seq anonymizers.

10 Domain Generalization: Transfer to Legal Text

A key practical question is whether anonymization models trained on general conversational data can *generalise* to domain-specific corpora. We evaluate both approaches on the Text Anonymization Benchmark (TAB; Pilán et al. 2022), comprising 500 annotated segments from European Court of Human Rights (ECHR) decisions. TAB presents a challenging out-of-domain test with formal legal prose and domain-specific entities entirely absent from our training data.

10.1 Zero-Shot Transfer

Both approaches were evaluated on the TAB test split *without fine-tuning*. We report entity recall against TAB gold annotations, Entity Leakage Rate (ELR), BERTScore F1, and ROUGE-L.

Approach 1 (Seq2Seq). Table 15 summarises seq2seq zero-shot performance. Models transfer well for high-frequency types (PERSON ≥ 0.95, CODE ≥ 0.97), but struggle with rare or legal-specific entities (ORGANIZATION and QUANTITY recall falls to 0.39–0.72). Crucially, *the Sample Leakage Rate is 0% for all models*: no passage passes through unchanged, confirming active rewriting on unseen domains.

Model	Recall	ELR↓	B-4	R-L	BS-F1
t5-eff-tiny	0.884	11.56	63.62	79.37	90.23
t5-small	0.869	13.14	70.46	83.88	90.80
flan-t5-small	0.874	12.61	69.21	82.83	90.66
bart-base	0.830	16.97	67.37	84.07	**91.08**

Table 15: Approach 1 zero-shot results on TAB. ELR = Entity Leakage Rate (%); B-4 = BLEU-4; R-L = ROUGE-L; BS-F1 = BERTScore F1.

Approach 2 (Pipeline). Table 16 shows pipeline zero-shot results. Entity recall (0.716–0.777) is notably lower than seq2seq models. The domain gap severely impacts LOCATION (0.03–0.09), ORGANIZATION (0.03–0.68), and NUMBER (0.08–0.24) recall, as these are under-represented in the AI4Privacy training distribution. Despite this, DeBERTa-MLM fillers consistently outperform BART fillers on quality metrics.

Masker	Filler	Recall	ELR↓	B-4	R-L	BS-F1
distilroberta	bart-base	0.777	18.43	62.13	78.28	88.63
distilroberta	deberta-mlm	0.777	21.31	78.10	90.57	93.27
roberta	bart-base	0.776	18.21	62.54	78.21	88.52
roberta	**deberta-mlm**	0.776	21.45	**78.41**	90.43	93.23
deberta	bart-base	0.716	24.12	63.08	78.73	88.99
deberta	deberta-mlm	0.716	27.36	79.03	**90.65**	**93.52**

Table 16: Approach 2 zero-shot results on TAB. Bold combo selected as seed for fine-tuning.

10.2 Fine-Tuning on TAB

Unlike the modular pipeline, Approach 1 requires paired (*original text*, *fluent anonymized text*) examples. Since TAB only provides entity-span annotations and categorical placeholders (e.g., `[PERSON]`), fine-tuning Approach 1 requires passage-level augmentation beyond the dataset's scope. Conversely, Approach 2's modularity allows direct fine-tuning: the masker on TAB's BIO-tagged spans, and the filler on context mask-fill pairs.

We fine-tuned our highest-performing zero-shot pipeline (**roberta** + **deberta-mlm**) on TAB training data (Table 17) to assess its adaptability.

Masker	Filler	Recall	ELR↓	R-L	BS-F1
roberta-tab	**deberta-tab**	**0.865**	**13.51**	**90.57**	**96.07**
(zero-shot reference)					
roberta	deberta-mlm	0.776	21.45	90.43	93.23

Table 17: Approach 2 after fine-tuning on TAB, demonstrating strong domain adaptation.

The **roberta-tab** + **deberta-tab** combination improves significantly: entity recall rises to **0.865**, ELR drops to **13.51%**, and BERTScore F1 reaches **96.07**. This is driven by massive recall gains in domain-specific types: LOCATION (0.03 → 0.84), ORGANIZATION (0.03 → 0.74), and NUMBER (0.08 → 0.70). These results directly validate that the modular mask-and-fill architecture can effectively generalize and be optimized for highly specialized, out-of-domain tasks (such as formal legal documents) through targeted fine-tuning.

11 Conclusion

We presented a comprehensive empirical study of neural PII anonymization, comparing direct seq2seq rewriting against modular mask-and-fill pipelines. BART-base (139M parameters) achieves the best overall performance (0.93% ELR, BERTScore 92.74), outperforming both larger distilled models and all pipeline combinations. While DeBERTa-MLM metrics nominally outperform BART in the mask-and-fill pipeline, qualitative inspection reveals this is an artifact of the metric: DeBERTa's inability to dynamically expand token lengths leads to degenerate, non-fluent outputs that artificially minimize Entity Leakage (ELR). Consequently, we conclude that unconstrained Seq2Seq architectures (like BART-base) are the architecturally correct choice for PII anonymization, as they preserve both privacy and genuine grammatical fluency. Both approaches transfer zero-shot to ECHR legal text with strong recall for person names, dates, and codes, but struggle with domain-specific organisational and location entities. Fine-tuning the pipeline on TAB reduces ELR from 21.45% to 13.51% and raises entity recall from 0.776 to 0.865 by substantially improving LOCATION, ORG, and NUMBER coverage.

Our security analysis reveals a stark vulnerability asymmetry: seq2seq outputs are moderately susceptible to model inversion (29% ERR), while pipeline outputs are highly vulnerable (∼77%), due to the explicit positional signals in placeholder-format text. Adversarial training reduces seq2seq inversion vulnerability by 98.6% without significant quality degradation.

Future Work. (1) Differentially private training (DP-SGD via Opacus) on the pipeline filler to provide formal (ε, δ)-DP guarantees. (2) Positional obfuscation of pipeline placeholders to reduce inversion vulnerability. (3) Scaling to BART-large and T5-base on cloud TPU. (4) Extending to multilingual PII (the AI4Privacy dataset covers 55+ languages).Although approach 2 is still multilingual by its nature we need to explore other languages for approach 1. (5) Fine-grained ablation of the PII-Aware Loss weight.

Limitations. (1) TAB fine-tuning was performed only for Approach 2 due to the paired-fiction requirement of Approach 1. (2) Adversarial hardening was applied only to BART-base; pipeline hardening remains an open problem. (3) Model outputs occasionally include corrupted tokens under adversarial pressure.

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Data and Model Availability

To support reproducibility and future work, all artifacts developed in this research are publicly

available on HuggingFace. We provide hyperlinked aliases below to access the respective repositories.

Released Datasets.

- **Anonymization:** Saha-AL (Human Annotated)
- **Adversarial Evaluation:** Adv-Sentences, Adv-Pipeline Robustness
- **Model Inversion:** Inversion-BART (Seq2Seq), Inversion-RoBERTa-BART, Inversion-RoBERTa-DeBERTa

Released Models.

- **Baseline & PII-Aware:** 10-Model Repository (5 each based on Standard CE and PII-Aware)
- **Pipeline Base:** DistilRoBERTa Masker, RoBERTa Masker, DeBERTa Encoder, DeBERTa Filler, BART Filler
- **TAB Fine-Tuned:** TAB-RoBERTa Masker, TAB-BART Filler, TAB-DeBERTa Filler
- **Model Inverters:** Inverter-BART (Seq2Seq), Inverter-RoBERTa-BART, Inverter-RoBERTa-DeBERTa
- **Adversarially Hardened:** Hardened-BART-Base (Seq2Seq), Hardened-BART-Filler (Pipeline), Hardened-DeBERTa-Filler (Pipeline)